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Class Test.

Computer vision
ITA0506

1) Image Enhancement is the process of improving the visual appearence of an image or making it more suitable for Analysis of by computer vision systems. In the spatial domain enhancement is performed by directly modifying the pixel values of the image. spatial domain.

Spatial domain methods are simple, computationally efficient & widely used in digital image processing. The three major techniques are gray-level transformation histogram processing & spatial filtering.

i) Gray-Level Transforming:

Gray level Transformation is a point processing technique in which each pixel value is changed indepdently according to predefing mathematical function.

- log Transformation.
- Gamma Transformation.
- Constrast Streching.

2) Histogram processing techniques:

A histogram represents the distribution of gray levels image. Histogram processing enhances image quality by redistributing pixel intensities.

- Histogram Equalization.
- Histogram specification.

3) Spatial filtering:

Spatial filtering modifies pixels values using a small neighbourhood called a filter or mask.

- Smoothing filters.
- Sharpening filters.

2) Image smoothing is the process of reducing noise & small variations in an image. It works by replacing each pixels value with average or weighted average of its neighbouring pixels. As a result sudden intensity changes are reduced producing a smoother image.

i) mean filter:

Replace the center pixels by average of the matrix.

ii) median filter:

Replaces the center pixel with the median values of neighbouring pixels. It is very efficient for removing salt & pepper noise while mean filter.

iii) Gaussian filter:

uses a weighted average based on the Gaussian distribution. pixels closer to the center recieve high weights & more natural image.

Image sharpening:

image sharpening is the process of highlighting edges and fine details by emphasing Rapid intensity changes b/w neighbouring pixels.
It improves the visibility of object boundaries.

- Laplacian Filter: uses the Second derivate to detect regions of rapid intensity changes.
- Prewitt Filter: Similar to the Sobel filter but a similar gradient calculation.

3) Image enhancement in frequency domain is a techique in which an image is processed by modifying its frequency components. & The image is converted back using the inverse Transform.

Low - pass Filter (LPF):

- Passes low frequency components.
- Remove noise.
- Produces a smoother image.
- Used in edge detection & features extraction.

High - pass Filter (HPF):

- Pass high frequency components.
- Enhances edges & fine details.
- Produces a Sharper image.
- Used in edges detection & feature extraction.

Low pass filter	High-pass filter.
Remove noise	Enhance edges.
Smooths image	Sharpens image.
Preserves low frequencies.	Preseves high frequencie
Used for denoising.	Used for features Enhancements.

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Image Segmentation divides an image into meaningful regions to simplify analysis & object recognition.

Technique	Working Principle	Advantages
Point Detection.	Detects isolated pixels.	Simple.
Line Detection.	Detects horizontal, vertical or diagonal lines.	Accurate boundaries.
Edge Detection.	Detects intensity changes.	Detects the image.
Thersholding	Seprates objects intensity values.	Fast and Simple.
Region - Based Segmentation.	Groups similar neighbouring pixels.	Produces complete Regions.

5) Edge Detection : Edge detection identifies significant intensity changes in an image, which represent object boundaries. Common operators using include Sobel, Prewitt, Canny, Laplacian.

5) Edge detection : Edge detection identifies significant intensity changes in an image, which represent object boundaries, common operators includes continous boundaries using pixel connectivity and edges. orientation.

Boundary Detection : Boundary detection extracts the outer counter of an object after segmentation enabling accurate shape Analysis.

- Improves object recognition.
- Helps identify defects in industrial inspection.
- Assists in medical image Analysis.
- Enables accurates image segmentation.
- Supports pattern recognition & computer vision. applications.

In medical imaging, boundary detection identifies tumors.

In manufacturing edge detection identifies cracks and defects.